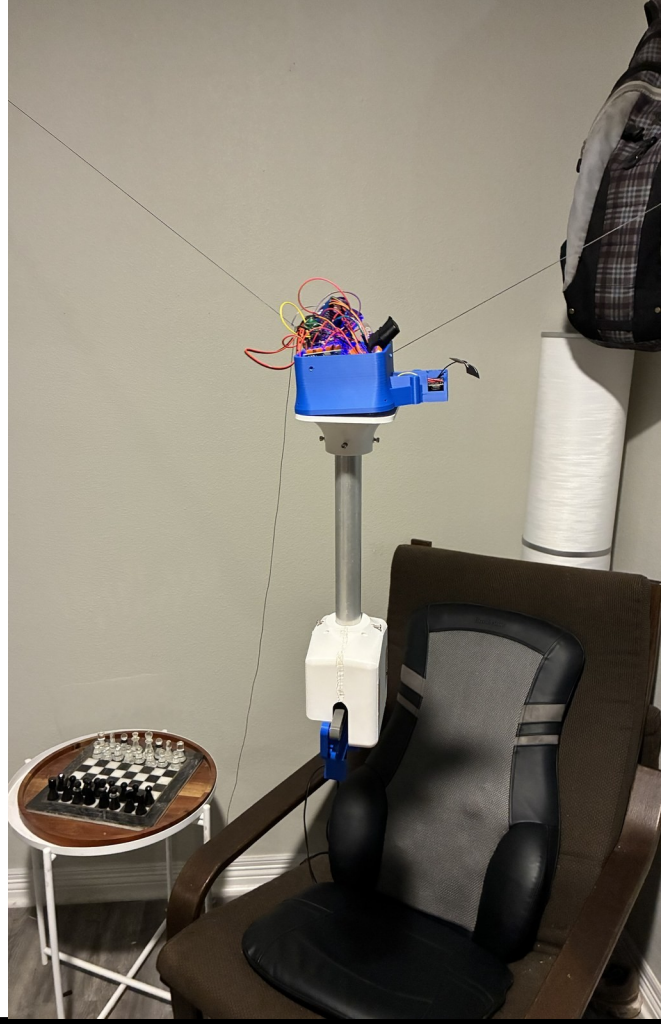
# RoomCleaner

A four-cable ceiling robot that picks laundry off the floor

Jasper Buntinx | Mechanical Engineering | Texas A&M

October 27, 2026



# About me

## Education: Texas A&M

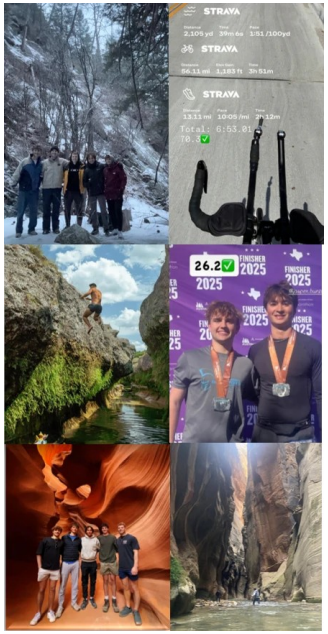
- Bachelor's, Mechanical Engineering, August 2027

## Experience

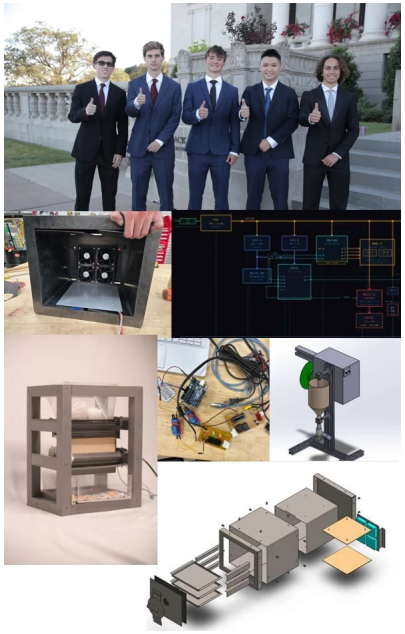
- ITP Lab, undergraduate research
- Rochester Sensors, mechanical intern
- Petrobras, mechanical intern
- Aggies Create, EcoFil electrical lead

## Hobbies

- Personal projects
- Soccer
- Running
- Bouldering
- And man, do I love hiking.




Outdoors and endurance



Engineering




Community

# Project scope

### The problem

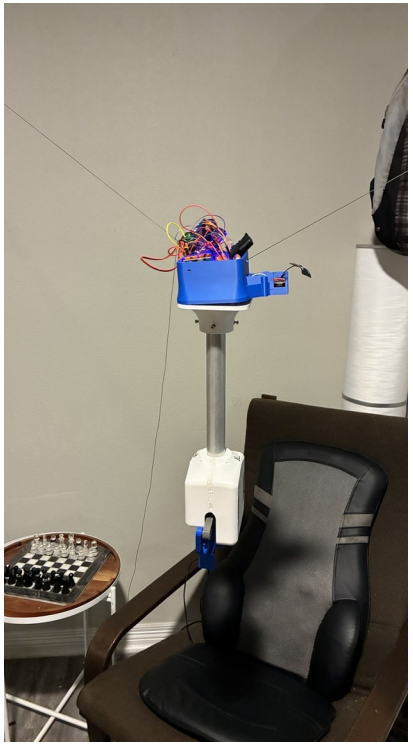
- Laundry on the floor of a 3.35 x 2.74 m room (as reported)
- Ceiling 2.74 m; four studs near the ceiling corners
- Design load: a pair of jeans, 0.9 kg (screening value)
- Plus 2 m/s² for stops and snags (assumed)

### The solution

- Four ceiling winches, one claw on four cables
- Nothing on the floor; the room is the frame
- A camera on the claw finds the laundry
- Floor to hamper with no human touch

### How we do it

- I approve each move on the phone
- PAUSE holds position with torque on
- The switched power strip is the emergency stop; it drops the claw
- First gate: a 1 lb test weight, then jeans
- Parts in the robot: about $211 for three stations, plus $38 filament (partly estimated)



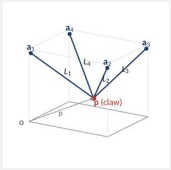
The claw hanging in the room corner (Oct 27)

# How cable length becomes position, and torque becomes pull

### Cable length → position

$$L_i = \|\mathbf{a}_i - \mathbf{p}\|$$

$\mathbf{a}_i$ anchor, $\mathbf{p}$ claw




- Line per spool turn = 2π·r: 2π·7.5 mm = 47.1 mm (empty) → 2π·10 mm = 62.8 mm (full) (calculated)
- Four cable lengths, three position unknowns: overdetermined, solved by least squares
- The camera on the floor board fixes the anchors and cable lengths at startup (calibration)
- Validated in the simulator and by held-out poses during calibration (simulated); no physical position measurement yet

### Torque → pull on the line

$$I = \frac{V_{\text{ref}}}{8\,R_s} \qquad \tau_{\text{hold}} = \tau_{\text{rated}}\,\frac{I}{I_{\text{rated}}}$$

$$\tau_{\text{run}} \approx 0.7\,\tau_{\text{hold}}\ \text{(assumed)} \qquad F = \frac{\tau}{r}$$

- At 1.5 A: τ_run = 0.7 × 0.36 N·m = 0.252 N·m; F = 0.252 / 0.0075 m = 34 N empty, 0.252 / 0.010 m = 25 N full (calculated)
- Validated: Vref measured on the board; the 0.7 running fraction is assumed; the pull test is planned

### Pull → what four cables hold

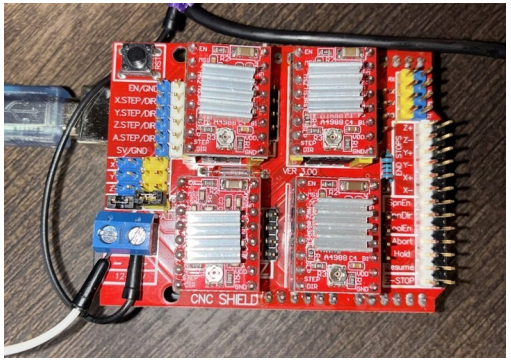
$$T = \frac{W}{4\sin\varphi} \qquad \sin\varphi = \frac{d}{\sqrt{d^2 + D_h^2}}$$

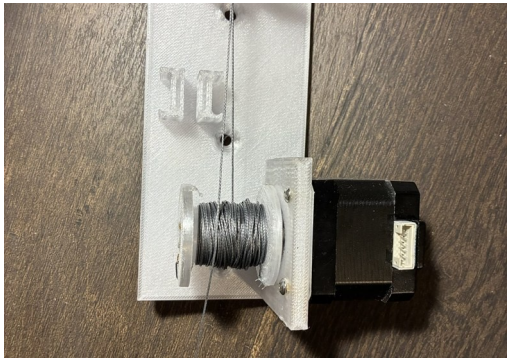
- W = (0.454 + 0.45 kg) × 9.81 = 8.86 N, D_h = 2.166 m; d = 0.15 m: 8.86 / (4 × 0.069) = 32 N; 16 / 8.3 / 4.6 N at 0.3 / 0.6 / 1.2 m (calculated)
- Shallow cable: T ≈ W·D_h / (4d), so halving the depth doubles the tension
- Why the planner keeps the claw at least 0.4 m below the anchors
- Validated: calculated only; no cable tension has been measured on the rig

# Generation 1: one central unit

### The first design

- One Uno and one CNC shield drove four corner motors
- Wires from that one box to every corner
- An overhead ceiling camera scanned the room
- A printed corner plate screwed into a joist: a hook is a pivot, a plate is not
- Each cable found zero against a bead at the outlet
- Bench: a station lifted a test weight; the cable math held (measured)





The central board; a gen-1 corner plate; a gen-1 station lifting a test weight

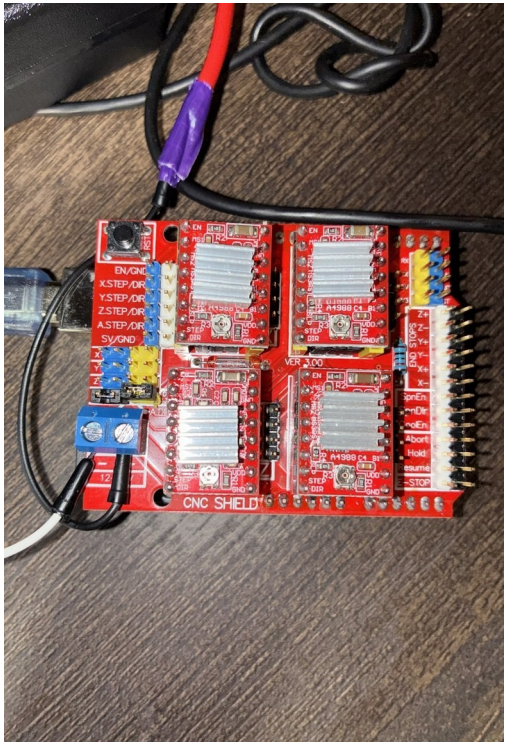
# Problems with Generation 1

### One central board

- Every motor and every sensor wired back to one place
- One board steps all four motors; a timing error skews the cables
- Skew $\Delta x = v \cdot \Delta t = 50$ mm/s × 0.05 s = 2.5 mm per 50 ms (calculated)
- Moving a winch means re-running wire to the box

### Ceiling camera and bead homing

- The ceiling camera: one more device to mount, wire and calibrate per room
- From above, the claw and cables hide the grasp spot
- Its floor map assumes a flat floor; retired Oct 21
- Bead homing fixes a minimum length on every cable: next slide



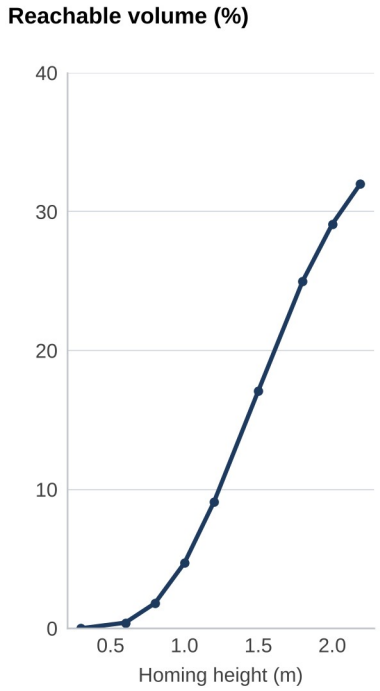

The central board: one shield, four drivers

# Homing: one floor board instead of four beads

## Bead homing

$$\text{reachable}(\mathbf{p}) \Leftrightarrow \|\mathbf{a}_i - \mathbf{p}\| \geq L_{i,\,\text{home}} \quad \text{for every } i$$

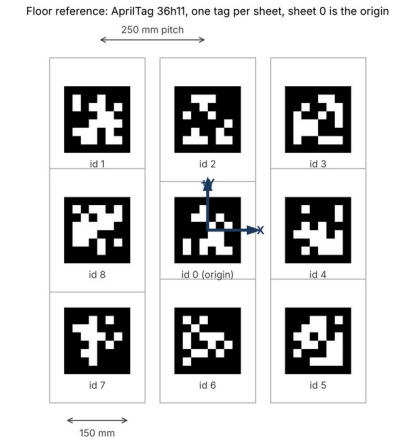
- A stopper bead cannot pass its outlet: after homing, no cable can be shorter than at home
- Homing all four at once fixes a minimum length on every cable at the same time
- Reachable volume, grid count (calculated): 0.0 % (home 0.3 m) to 32.0 % (home 2.19 m); 4.7 % at 1.0 m, 29.1 % at 2.0 m




Grid count of the condition above

## Floor-board homing

- The camera measures the claw pose against tag 0
- Cable lengths follow from L_i = ‖a_i − p‖
- No length is clamped: the whole working volume stays available
- Nine sheets, 150 mm tags, 250 mm pitch, id 0 = origin



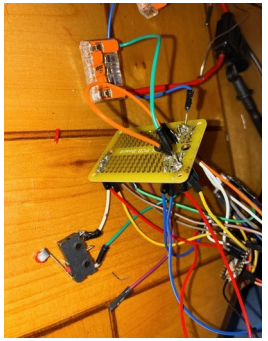
# What a winch on a wall needs to work

### The switch (button)

- Things go wrong: a camera fails, a spool overwinds, tension rips a mount out, a motor stalls
- A switch the cable's bead can hit: that station stops
- Fail-safe wiring: LOW at rest, HIGH pressed or with a broken wire

### The encoder

- On the spool: measures the discrepancies while testing; data first
- Confirms every commanded step
- Catches slip: the tension sensor
- If the camera proves good enough, it can go later



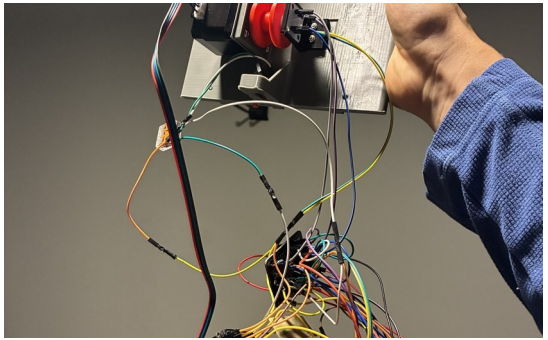
Homing switch



Switch pressed into the slide-in

### And now all this wiring, from one box

- Motor, switch and encoder in every corner: 10 conductors per corner
- Routes 6 + 21 + 23 + 32 = 82 ft; × 10 conductors = 820 ft (250 m) of wire to one box (calculated)



One slide-in, its harness hanging

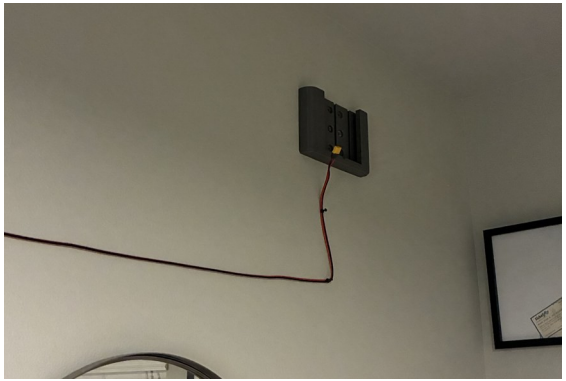
# The solution: a modular box that only needs power

### A box with its own brain

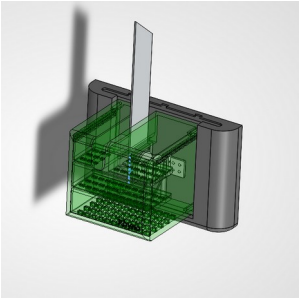
- An ESP32 in each box, with its driver, buck, switch and encoder
- The only thing going in is power: one 12 V pair
- 82 ft × 2 conductors = 164 ft (50 m) of wire, not 820 ft (calculated)
- Power can come from anywhere, so a box sits in any position and orientation
- The room's geometry no longer limits where the winches go

### Wire drop: Gen 1 leads vs one power pair

- $V = I \cdot \rho \cdot 2L/A$; copper $\rho = 1.724 \times 10^{-8}$ Ω·m, L = 9.75 m (32 ft) one way, I = 2 A
- 24 AWG Gen 1 motor leads, A = 0.205 mm²: R = 1.64 Ω, 3.29 V (27 % of 12 V), 6.6 W (calculated)
- 16 AWG power pair, A = 1.309 mm²: R = 0.257 Ω, 0.51 V (4.3 %), 1.0 W (calculated)
- Thin motor leads over a room-length run lose a quarter of the supply; one thick power pair loses 4 %
- 16 AWG: $30 per 100 ft × 65 ft = $19.50 (price estimated)



The wall mount: one power plug



The station: board, driver, motor, fin (CAD)

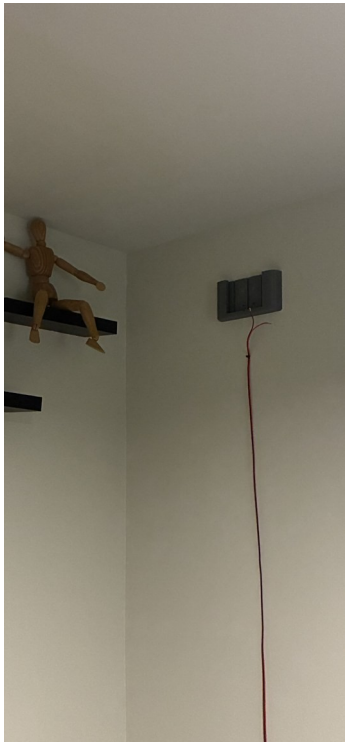
# Modular, for the everyday user

### Unmount, remount, plug in

- A box slides off its wall mount and unplugs
- Guests coming: each box goes in a drawer
- Back on: one plug, it joins the Wi-Fi by itself

### No screwdriver on the wall

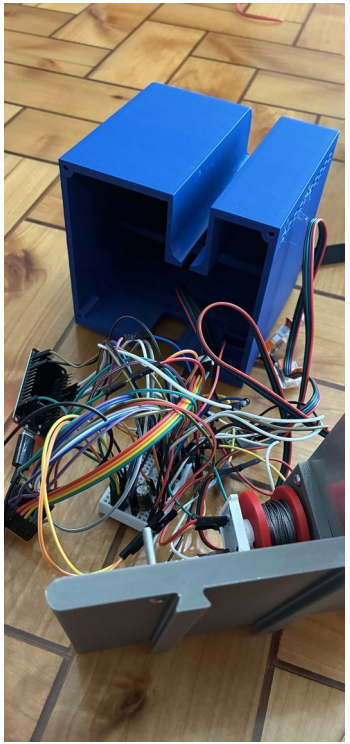
- Slip the station out of the mount, open the electronics
- Close it, slide it back: the mount and its screws stay in the stud
- The mount is separate: a stud today, a pole clamp later



The wall mount alone, lead hanging



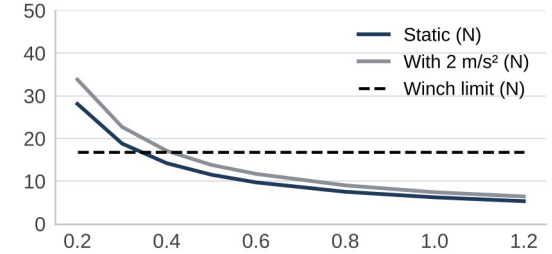

The station slid onto it



Shell open on the bench

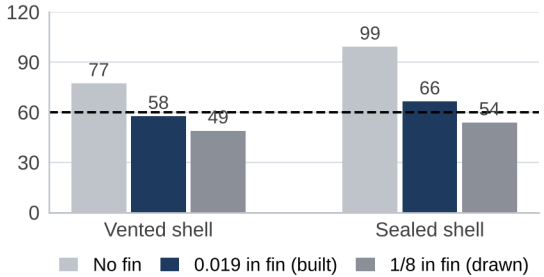
# Trade-offs: what each component needs

**Tension per cable (N) vs depth (m)**



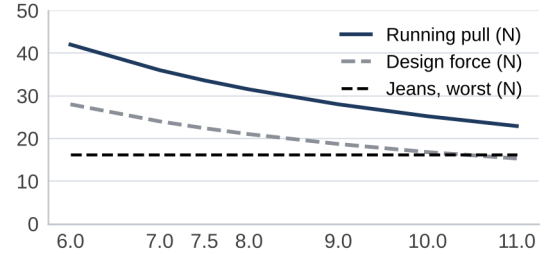

**T = m(g + a) / (4 sin φ): 1 lb at d 0.4 m → 17.1 N vs 16.8 N limit; jeans need d ≥ 0.6 m (16.6 N)**

**Motor case (°C) at 1.5 A; PETG limit 60 °C**



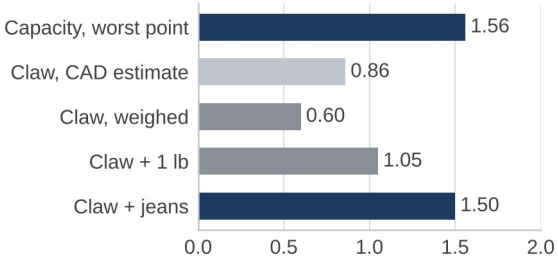

**Need < 60 °C. Two-node thermal network; input P = I²R = 1.5² × 1.8 Ω = 4.05 W**

**Pull per winch (N) vs spool radius (mm), 1.5 A**



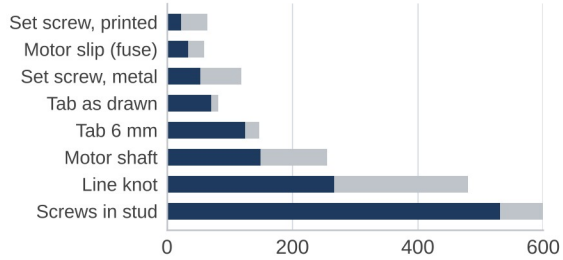

**F = 0.252 N·m / r: 34 N (7.5 mm), 25 N (10 mm); ÷ 1.5 = 16.8 N ≥ 16.2 N jeans**

**Winches can hold vs must hold (kg)**




**16.8 N / (1.101 × 9.81) = 1.56 kg vs 1.50 kg: +4 %; claw 0.60 kg weighed vs 0.45 budget**

**Tension at which each part gives (N)**




**Motor slip 0.36 N·m / 7.5 mm = 48 N; × 1/√2 = 34 N, × 1.11 / cos 26° = 59 N**

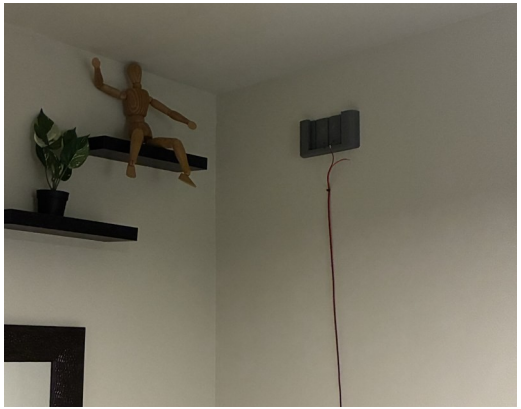
## The factors

- Pull ∝ I; heat P = I²R ∝ I²
- 1.0 → 1.5 A: pull × 1.5, heat × 1.5² = 2.25 (calculated)
- Every gram on the claw is a gram of payload
- Printed set screw (22–64 N) may slip first: metal insert

# The mount assembly: slide-in on a dovetail

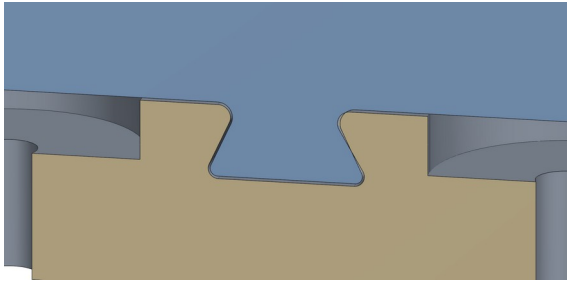
### Slide-in mount and load path

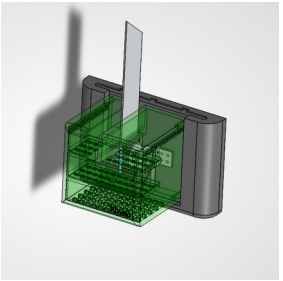
- The wall mount screws into a stud; the station slides onto it on a dovetail
- Dovetail: wall mount offset outward 0.175 mm per side; clearance set from fit coupons
- Load path: cable, spool, set screw, motor, tab, mount, screws, stud
- Motor slips first: F = τ/r = 0.36 N·m / 7.5 mm = 48 N; 34–59 N over torque vector and cable angle (calculated)
- The spool encoder sees the slip: tracking fault, every station holds






The mount alone, before the station slides in; the station on the mount





Rib in the dovetail groove (CAD); the station with its shell transparent (CAD)

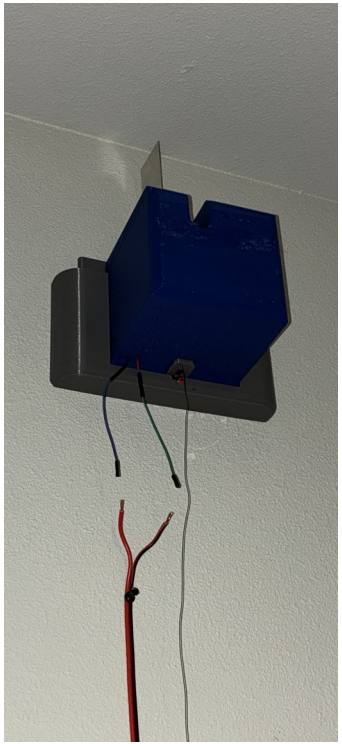
# The mount assembly: fin, vents and the wall

### Fin, vents, stud

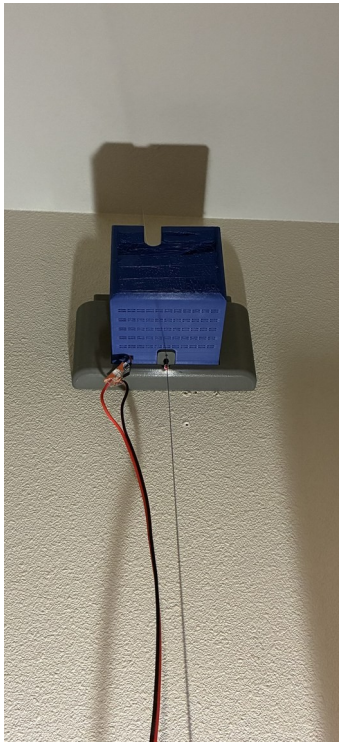
- The fin: aluminium on the motor face, out through a slot in the top of the shell
- Motor heat $P = I^2R = 1.5^2 \times 1.8\ \Omega = 4.05$ W leaves through vents and fin (calculated)
- First station in a stud on the TV wall, 12 V down the wall
- The slide-in binds on insertion today; a 1° draft on the dovetail is the next iteration



On the TV wall, 12 V leads down



From below: fin through the slot



The line leaves the switch tab straight down

# Manufacturing: in-house processes and fit control

## Processes

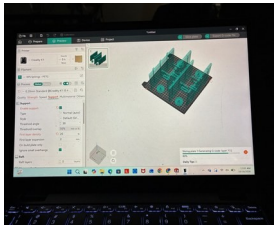
- PETG, FDM: 0.4 mm nozzle, 0.20 mm layers, supports at 30°; loads along the layers
- Aluminium fin: sheared blank, five holes drilled in one mill setup
- First hole is the mill zero: pattern ±0.1 mm vs ±0.5 mm sheared edges
- Heat-set inserts: M3 gear housing, M2.5 encoder board

## Make vs buy

- Bought: steppers, drivers, boards, servo, camera, battery, encoders
- Made: every printed part and the fin
- Filament per station: 0.762 kg × $15/kg = $11.43 (estimated)
- Robot: $210.78 parts + $37.74 filament, three stations (partly estimated)

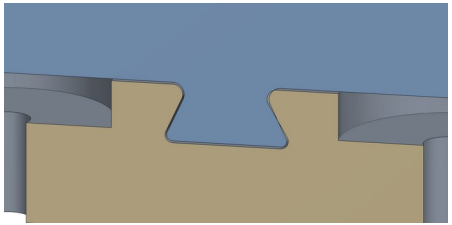
## Fits

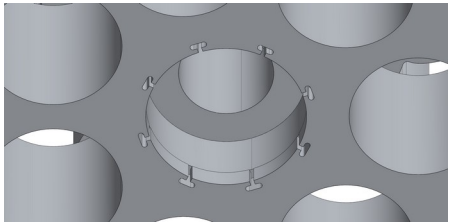
- Printed holes come out undersize: every printed fit carries an allowance
- Spool bore: Ø5.2 printed too tight on the Ø5.0 shaft; opened to Ø5.45
- Dovetail: wall mount offset outward 0.175 mm per side for clearance, 30° flanks
- Bearing pocket Ø14.075 behind a Ø14.000 entry lip, 8 T-slots
- Fit coupons printed first; the allowances above came from them (measured)
- A failed layer was kept on one shell only: the shell carries no load
- TPU 95A grip pads: next iteration





Slicer (K1, PETG); drilling the fin





Dovetail rib; bearing pocket and lip (CAD)

# The claw, and why it is modular

### The claw today

- Four-bar hands on a geared crank; 53 mm opening (CAD)
- A camera bay in the electronics housing; the camera looks down the tube
- One 7.4 V two-cell pack: 6 V regulator for the servo, 5 V buck for the board
- Weighed 0.60 kg with battery against a 0.45 kg budget

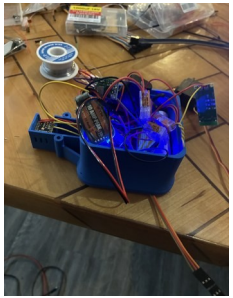
### Modular, on purpose

- The gripper attaches at the tube, so other grippers can be tried
- A hard pinch was expected to grip better; testing showed alignment problems
- The first rack-and-pinion jaws left a 2 mm gap on thin cloth (Sep 30)
- TPU grip pads go on in the next iteration
- The v6 clamp was 4.6 x its mass budget in CAD: mass is tracked per part since









Assembled; the hands (CAD); the v6 clamp (CAD); the housing open

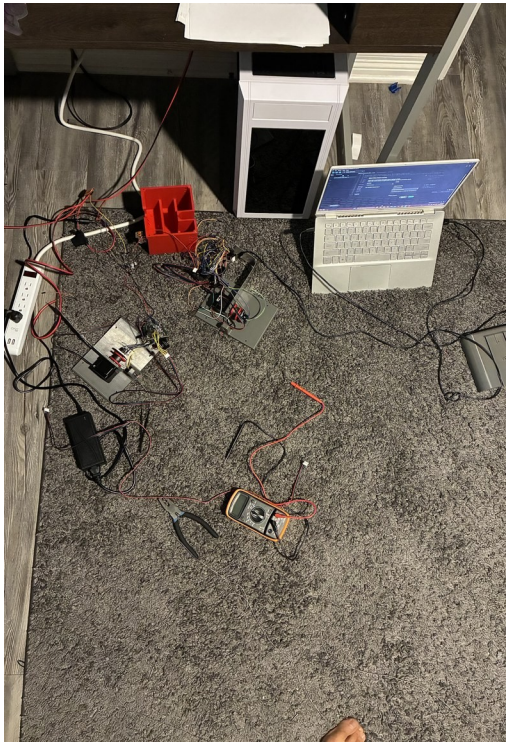
# Electrical and testing: bench, then wall

### Bench

- Station boards: bench checklist passed on both controller boards (measured)
- Claw: camera live view, battery sense and servo pass (measured)

### On the wall

- A boxed station boots and joins Wi-Fi by itself
- Firmware updates over Wi-Fi; no USB cable after boxing
- Motors stay off until the phone arms them; updates only while disarmed
- Planned: pull test, hold test, first three-station lift



Bench: two stations, laptop, meter, 12 V supply



The first station on the wall

# Startup and self-calibration: six states before the robot is READY

### 1 SELF_CHECK

- Claw online, battery above minimum
- Camera confirms the claw upright at the known height on the floor board
- Stations healthy, homing switches closed; arm

### 2 SLACK_TAKEUP

- One cable at a time reeled in until the camera sees the claw slide or tip (onset)
- Records spool position and slide bearing; backs off
- Bearings checked against the anchor layout; first fit of cable lengths

### 3 INITIAL_LIFT

- Pretension all cables; lift in small packets
- Abort on lateral drift, tilt, or commanded lift exceeding observed rise (slack cable)

### 4 VISUAL_INITIALIZATION

- Settled frame batch of the floor board
- Refit cable lengths and the room-to-marker transform

### 5 CALIBRATION

- Local → intermediate → expanded pose sets chosen by D-optimal design; refit between stages
- Held-out witness poses
- Accept on witness residuals, identifiability, predicted length sigma over the work volume

### 6 REFERENCE_VERIFICATION

- Optional bead references
- Trust level VERIFIED or LOCAL_ONLY from the margin check

**No automatic retry; any station reboot forces the full sequence again.**

# Next iteration: what was not working, and what changes

### 3:1 printed gear stage

- 18T / 54T spur, printed PETG pinion
- Designed and checked in CAD; pinion printed; not fitted
- Pull × 3 × 0.9 (η) = 2.7 at equal current (calculated)
- At 1.0 A: heat × $(1.0/1.5)^2 = 0.44$ (calculated)

### 1° draft on the dovetail

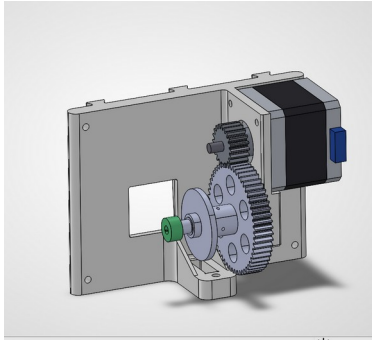
- Today the slide-in binds on insertion
- Thicker at the top, thinning toward the bottom
- The station self-locates as it slides in (planned)

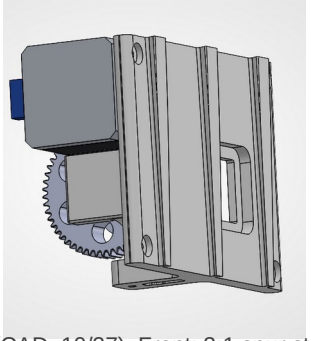
### Lighter back mount

- Today: 597 $cm^3$ solid, far stronger than the motor can pull
- Thinner walls and ribs: about 200 $cm^3$ (planned)
- Less print time and filament

### Also next / still planned

- TPU grip pads on the jaws (planned)
- Pull test at 1.5 A: the measured limit goes into the planner (planned)
- Hold test: 90 minutes at 1.5 A, motor below 60 °C (planned)
- First three-station lift: self-check, calibration, a short move with 1 lb (planned)





Next iteration (CAD, 10/27). Front: 3:1 spur stage, spool on its own shaft, encoder magnet cap. Back: slide-in rails, window, screw holes

# Thank you

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